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COHORT C

Query:
CREATE database Projects;

Query:
CREATE TABLE Department(deptNo VARCHAR(30) PRIMARY KEY, deptName VARCHAR(30) NOT NULL);

Results:

Field	Type	Null	Key	Default	Extra
deptNo	varchar(30)	NO	PRI	NULL	
deptName	varchar(30)	NO		NULL	

3 rows in set (0.042 sec)

Query:
CREATE TABLE Employee(empNo VARCHAR(5) PRIMARY KEY, fName VARCHAR(30) NOT NULL, lName VARCHAR(30) NOT NULL, address VARCHAR(70), DOB DATE NOT NULL, sex CHAR(1) NOT NULL CHECK (sex IN ('M', 'F')), position VARCHAR(30) CHECK(position IN ('Team Leader', 'Analyst', 'Software Developer')), deptNo VARCHAR(30) NOT NULL, FOREIGN KEY (deptNo) REFERENCES Department(deptNo));

Results:

Field	Type	Null	Key	Default	Extra
empNo	varchar(5)	NO	PRI	NULL	
fName	varchar(30)	NO		NULL	
lName	varchar(30)	NO		NULL	
address	varchar(70)	YES		NULL	
DOB	date	NO		NULL	
sex	char(1)	NO		NULL	
position	varchar(30)	YES		NULL	
deptNo	varchar(30)	NO	MUL	NULL	

8 rows in set (0.031 sec)

Query:
CREATE TABLE Project(projNo VARCHAR(30) PRIMARY KEY, projName VARCHAR(30), deptNo VARCHAR(30), FOREIGN KEY (deptNo) REFERENCES Department(deptNo));

Results:

Field	Type	Null	Key	Default	Extra
projNo	varchar(30)	NO	PRI	NULL	
projName	varchar(30)	YES		NULL	
deptNo	varchar(30)	YES	MUL	NULL	

3 rows in set (0.034 sec)

Query:
CREATE TABLE WorksOn(empNo VARCHAR(5), projNo VARCHAR(30), dateWorked DATE, hoursWorked int CHECK(hoursWorked >= 0 AND hoursWorked <= 40), PRIMARY KEY (empNo, projNo, dateWorked), FOREIGN KEY (empNo) REFERENCES Employee(empNo), FOREIGN KEY (projNo) REFERENCES Project(projNo));

Results:

Field	Type	Null	Key	Default	Extra
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Field	Type	Null	Key	Default	Extra
empNo	varchar(5)	NO	PRI	NULL	
projNo	varchar(30)	NO	PRI	NULL	
dateWorked	date	NO	PRI	NULL	
hoursWorked	int(11)	YES		NULL	

4 rows in set (0.035 sec)

//creating tables

Query:

```
INSERT INTO Department(deptNo, deptName) VALUES
('D001', 'Software Development'),
('D002', 'Business Analysis'),
('D003', 'Human Resources'),
('D004', 'Finance'),
('D005', 'Marketing'),
('D006', 'Research'),
('D007', 'Cyber Security'),
('D008', 'Networking'),
('D009', 'Operations'),
('D010', 'Customer Support');
```

Result:

deptNo	deptName
D001	Software Development
D002	Business Analysis
D003	Human Resources
D004	Finance
D005	Marketing
D006	Research
D007	Cyber Security
D008	Networking
D009	Operations
D010	Customer Support

10 rows in set (0.002 sec)

```
INSERT INTO Employee(empNo, fName, lName, address, DOB, sex, position, deptNo)
VALUES
('E1', 'Ama', 'Mensah', 'Accra', '1990-05-12', 'F', 'Team Leader', 'D001'),
('E2', 'Kwame', 'Asante', 'Kumasi', '1988-11-20', 'M', 'Analyst', 'D001'),
('E3', 'Abena', 'Owusu', 'Takoradi', '1992-07-15', 'F', 'Software Developer',
'D002'),
('E4', 'Kojo', 'Boateng', 'Cape Coast', '1991-03-10', 'M', 'Team Leader',
'D003'),
('E5', 'Akosua', 'Addo', 'Sunyani', '1993-08-25', 'F', 'Analyst', 'D004'),
('E6', 'Yaw', 'Darko', 'Tamale', '1989-01-05', 'M', 'Software Developer',
'D005'),
('E7', 'Efua', 'Agyeman', 'Ho', '1994-06-18', 'F', 'Team Leader', 'D006'),
('E8', 'Kofi', 'Ofori', 'Koforidua', '1990-09-30', 'M', 'Analyst', 'D007'),
('E9', 'Adwoa', 'Nyarko', 'Tema', '1995-12-14', 'F', 'Software Developer',
'D008'),
('E10', 'Nana', 'Tetteh', 'Accra', '1987-04-22', 'M', 'Team Leader', 'D009'),
('E11', 'Joseph', 'Arthur', 'Accra', '1994-02-15', 'M', 'Analyst', 'D002'),
('E12', 'Linda', 'Kusi', 'Kumasi', '1996-07-21', 'F', 'Software Developer',
'D003'),
('E13', 'Michael', 'Badu', 'Takoradi', '1991-11-10', 'M', 'Software Developer',
'D004'),
('E14', 'Patricia', 'Osei', 'Cape Coast', '1995-04-18', 'F', 'Analyst', 'D005'),
('E15', 'Daniel', 'Amoah', 'Tema', '1993-09-05', 'M', 'Software Developer',
```

```
'D006'),
('E16', 'Grace', 'Antwi', 'Ho', '1997-01-27', 'F', 'Analyst', 'D007'),
('E17', 'Samuel', 'Frimpong', 'Sunyani', '1992-06-12', 'M', 'Software
Developer', 'D008'),
('E18', 'Janet', 'Owusu', 'Tamale', '1994-12-03', 'F', 'Analyst', 'D009');
```

Results:

```
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+
| empNo | fName  | lName  | address | DOB      | sex | position |
| deptNo |
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+
| E1     | Ama    | Mensah | Accra    | 1990-05-12 | F   | Team Leader |
| D001   |
| E10    | Nana    | Tetteh | Accra    | 1987-04-22 | M   | Team Leader |
| D009   |
| E11    | Joseph  | Arthur | Accra    | 1994-02-15 | M   | Analyst      |
| D002   |
| E12    | Linda   | Kusi   | Kumasi   | 1996-07-21 | F   | Software     |
Developer | D003   |
| E13    | Michael | Badu   | Takoradi | 1991-11-10 | M   | Software     |
Developer | D004   |
| E14    | Patricia | Osei   | Cape Coast | 1995-04-18 | F   | Analyst      |
| D005   |
| E15    | Daniel  | Amoah  | Tema     | 1993-09-05 | M   | Software     |
Developer | D006   |
| E16    | Grace   | Antwi  | Ho       | 1997-01-27 | F   | Analyst      |
| D007   |
| E17    | Samuel  | Frimpong | Sunyani  | 1992-06-12 | M   | Software     |
Developer | D008   |
| E18    | Janet   | Owusu  | Tamale   | 1994-12-03 | F   | Analyst      |
| D009   |
| E2     | Kwame   | Asante | Kumasi   | 1988-11-20 | M   | Analyst      |
| D001   |
| E3     | Abena   | Owusu  | Takoradi | 1992-07-15 | F   | Software     |
Developer | D002   |
| E4     | Kojo    | Boateng | Cape Coast | 1991-03-10 | M   | Team Leader  |
| D003   |
| E5     | Akosua  | Addo   | Sunyani  | 1993-08-25 | F   | Analyst      |
| D004   |
| E6     | Yaw     | Darko  | Tamale   | 1989-01-05 | M   | Software     |
Developer | D005   |
| E7     | Efua    | Agyeman | Ho       | 1994-06-18 | F   | Team Leader  |
| D006   |
| E8     | Kofi    | Ofori  | Koforidua | 1990-09-30 | M   | Analyst      |
| D007   |
| E9     | Adwoa   | Nyarko | Tema     | 1995-12-14 | F   | Software     |
Developer | D008   |
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+
18 rows in set (0.001 sec)
```

Query:

```
ALTER TABLE Department ADD mgrEmpNo CHAR(5), ADD CONSTRAINT fk_manager FOREIGN
KEY (mgrEmpNo) REFERENCES Employee(empNo);
```

Results:

```
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| deptNo     | varchar(30)   | NO   | PRI | NULL    |       |
| deptName   | varchar(30)   | NO   |     | NULL    |       |
| mgrEmpNo   | char(5)       | YES  | MUL | NULL    |       |
```

```
+-----+-----+-----+-----+-----+
3 rows in set (0.035 sec)
```

Query:

```
UPDATE Department SET mgrEmpNo='E1' WHERE deptNo='D001';
UPDATE Department SET mgrEmpNo='E3' WHERE deptNo='D002';
UPDATE Department SET mgrEmpNo='E4' WHERE deptNo='D003';
UPDATE Department SET mgrEmpNo='E5' WHERE deptNo='D004';
UPDATE Department SET mgrEmpNo='E6' WHERE deptNo='D005';
UPDATE Department SET mgrEmpNo='E7' WHERE deptNo='D006';
UPDATE Department SET mgrEmpNo='E8' WHERE deptNo='D007';
UPDATE Department SET mgrEmpNo='E9' WHERE deptNo='D008';
UPDATE Department SET mgrEmpNo='E10' WHERE deptNo='D009';
UPDATE Department SET mgrEmpNo='E2' WHERE deptNo='D010';
```

Result:

```
+-----+-----+-----+
| deptNo | deptName          | mgrEmpNo |
+-----+-----+-----+
| D001   | Software Development | E1       |
| D002   | Business Analysis   | E3       |
| D003   | Human Resources     | E4       |
| D004   | Finance             | E5       |
| D005   | Marketing           | E6       |
| D006   | Research            | E7       |
| D007   | Cyber Security      | E8       |
| D008   | Networking          | E9       |
| D009   | Operations          | E10      |
| D010   | Customer Support    | E2       |
+-----+-----+-----+
10 rows in set (0.002 sec)
```

Query:

```
INSERT INTO Project(projNo, projName, deptNo) VALUES
('P001', 'Payroll System', 'D001'),
('P002', 'Sales Tracker', 'D002'),
('P003', 'HR Portal', 'D003'),
('P004', 'Budget Manager', 'D004'),
('P005', 'Market Survey', 'D005'),
('P006', 'AI Research', 'D006'),
('P007', 'Security Audit', 'D007'),
('P008', 'Network Upgrade', 'D008'),
('P009', 'Operations Dashboard', 'D009'),
('P010', 'Help Desk App', 'D010');
```

Result:

MariaDB [Projects]> select * from Project;

```
+-----+-----+-----+
| projNo | projName          | deptNo |
+-----+-----+-----+
| P001   | Payroll System    | D001   |
| P002   | Sales Tracker     | D002   |
| P003   | HR Portal         | D003   |
| P004   | Budget Manager    | D004   |
| P005   | Market Survey     | D005   |
| P006   | AI Research       | D006   |
| P007   | Security Audit    | D007   |
| P008   | Network Upgrade   | D008   |
| P009   | Operations Dashboard | D009   |
| P010   | Help Desk App     | D010   |
+-----+-----+-----+
10 rows in set (0.002 sec)
```

Query:

```

INSERT INTO WorksOn(empNo, projNo, dateWorked, hoursWorked) VALUES
('E1', 'P001', '2026-06-01', 8),
('E2', 'P001', '2026-06-02', 6),
('E3', 'P002', '2026-06-03', 7),
('E4', 'P003', '2026-06-04', 8),
('E5', 'P004', '2026-06-05', 5),
('E6', 'P005', '2026-06-06', 6),
('E7', 'P006', '2026-06-07', 8),
('E8', 'P007', '2026-06-08', 7),
('E9', 'P008', '2026-06-09', 6),
('E10', 'P009', '2026-06-10', 8),
('E11', 'P002', '2026-06-11', 8),
('E12', 'P003', '2026-06-12', 6),
('E13', 'P004', '2026-06-13', 7),
('E14', 'P005', '2026-06-14', 5),
('E15', 'P006', '2026-06-15', 8),
('E16', 'P007', '2026-06-16', 6),
('E17', 'P008', '2026-06-17', 7),
('E18', 'P009', '2026-06-18', 8);

```

Result:

empNo	projNo	dateWorked	hoursWorked
E1	P001	2026-06-01	8
E10	P009	2026-06-10	8
E11	P002	2026-06-11	8
E12	P003	2026-06-12	6
E13	P004	2026-06-13	7
E14	P005	2026-06-14	5
E15	P006	2026-06-15	8
E16	P007	2026-06-16	6
E17	P008	2026-06-17	7
E18	P009	2026-06-18	8
E2	P001	2026-06-02	6
E3	P002	2026-06-03	7
E4	P003	2026-06-04	8
E5	P004	2026-06-05	5
E6	P005	2026-06-06	6
E7	P006	2026-06-07	8
E8	P007	2026-06-08	7
E9	P008	2026-06-09	6

Question 7.25:

```

CREATE VIEW FemaleManagedProjects AS SELECT p.projName, p.projNo FROM Project p
JOIN Department d ON d.deptNo = p.deptNo JOIN Employee e ON d.mgrEmpNo = e.empNo
WHERE e.sex = 'F';

```

```

SELECT * FROM FemaleManagedProjects ORDER BY projNo;

```

Results:

projName	projNo
Payroll System	P001
Sales Tracker	P002
Budget Manager	P004
AI Research	P006
Network Upgrade	P008

5 rows in set (0.013 sec)

Question 7.26:

```
CREATE VIEW EmployeeInfo AS SELECT e.empNo, e.fName, e.lName, p.projName,
w.hoursWorked FROM Employee e JOIN WorksOn w ON e.empNo = w.empNo JOIN Project p
ON w.projNo = p.projNo;
```

Results:

empNo	fName	lName	projName	hoursWorked
E1	Ama	Mensah	Payroll System	8
E2	Kwame	Asante	Payroll System	6
E11	Joseph	Arthur	Sales Tracker	8
E3	Abena	Owusu	Sales Tracker	7
E12	Linda	Kusi	HR Portal	6
E4	Kojo	Boateng	HR Portal	8
E13	Michael	Badu	Budget Manager	7
E5	Akosua	Addo	Budget Manager	5
E14	Patricia	Osei	Market Survey	5
E6	Yaw	Darko	Market Survey	6
E15	Daniel	Amoah	AI Research	8
E7	Efua	Agyeman	AI Research	8
E16	Grace	Antwi	Security Audit	6
E8	Kofi	Ofori	Security Audit	7
E17	Samuel	Frimpong	Network Upgrade	7
E9	Adwoa	Nyarko	Network Upgrade	6
E10	Nana	Tetteh	Operations Dashboard	8
E18	Janet	Owusu	Operations Dashboard	8

18 rows in set (0.003 sec)

Question 7.27:

```
CREATE VIEW EmpProject(empNo, projNo, totalHours) AS SELECT w.empNo, w.projNo,
SUM(hoursWorked) FROM Employee e, Project p, WorksOn w WHERE e.empNo = w.empNo
AND p.projNo = w.projNo GROUP BY w.empNo, w.projNo;
```

Question 7.27a.:

```
SELECT * FROM EmpProject;
```

Result:

empNo	projNo	totalHours
E1	P001	8
E10	P009	8
E11	P002	8
E12	P003	6
E13	P004	7
E14	P005	5
E15	P006	8
E16	P007	6
E17	P008	7
E18	P009	8
E2	P001	6
E3	P002	7
E4	P003	8
E5	P004	5
E6	P005	6
E7	P006	8
E8	P007	7
E9	P008	6

18 rows in set (0.016 sec)

Question 7.27b.:

```
SELECT projNo FROM EmpProject WHERE projNo = 'SCCS';
```

Result:

Empty set (0.011 sec)

Question 7.27c.:

```
SELECT COUNT(projNo) FROM EmpProject WHERE empNo = 'E1';
```

Result:

```
+-----+
| COUNT(projNo) |
+-----+
|              1 |
+-----+
1 row in set (0.011 sec)
```

Question 7.27d.:

```
SELECT empNo, SUM(totalHours) AS totalHours FROM EmpProject GROUP BY empNo;
```

Result:

```
+-----+-----+
| empNo | totalHours |
+-----+-----+
| E1    | 8          |
| E10   | 8          |
| E11   | 8          |
| E12   | 6          |
| E13   | 7          |
| E14   | 5          |
| E15   | 8          |
| E16   | 6          |
| E17   | 7          |
| E18   | 8          |
| E2    | 6          |
| E3    | 7          |
| E4    | 8          |
| E5    | 5          |
| E6    | 6          |
| E7    | 8          |
| E8    | 7          |
| E9    | 6          |
+-----+-----+
18 rows in set (0.003 sec)
```

Question (General):

Query:

```
CREATE TABLE Part(partNo VARCHAR(10), contract VARCHAR(15), partCost
DECIMAL(10,2), PRIMARY KEY(partNo, contract));
```

Result:

```
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| partNo | varchar(10)   | NO   | PRI | NULL    |       |
| contract | varchar(15)  | NO   | PRI | NULL    |       |
| partCost | decimal(10,2) | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.032 sec)
```

Query:

```
INSERT INTO Part(partNo, contract, partCost) VALUES
('P001', 'C001', 850.50),
('P002', 'C001', 1250.75),
('P003', 'C001', 1950.00),
```

```
( 'P004', 'C002', 980.40),
( 'P005', 'C002', 1450.99),
( 'P006', 'C003', 1750.00),
( 'P007', 'C003', 1600.25),
( 'P008', 'C004', 275.80),
( 'P009', 'C005', 1999.99),
( 'P010', 'C005', 1850.30);
```

Result:

partNo	contract	partCost
P001	C001	850.50
P002	C001	1250.75
P003	C001	1950.00
P004	C002	980.40
P005	C002	1450.99
P006	C003	1750.00
P007	C003	1600.25
P008	C004	275.80
P009	C005	1999.99
P010	C005	1850.30

10 rows in set (0.001 sec)

Query:

```
CREATE VIEW ExpensiveParts(partNo) AS SELECT DISTINCT partNo FROM part WHERE
partCost>1000;
```

Result:

partNo
P002
P003
P005
P006
P007
P009
P010

7 rows in set (0.003 sec)

Discussion:

A materialized view stores the result of the query, so it must be updated whenever the Part table changes. If a new row is inserted with a partCost greater than 1000, the partNo should be added to the view. If a part's cost is updated, the view should also be updated by adding the partNo if the cost becomes greater than 1000, or removing it if the part no longer has any contract with a cost above 1000. Similarly, when a row is deleted, the partNo should only be removed from the view if there are no other contracts for that part with a cost greater than 1000.

The view can be maintained without accessing the Part table only when it is certain that the change cannot affect other rows. For example, if a new row with partCost greater than 1000 is inserted, the partNo can simply be added to the view. However, if a row with partCost greater than 1000 is updated or deleted, the Part table must be checked to determine whether another contract for the same partNo still has a cost above 1000 before removing it from the view.